

Codename One: A Cross-Device Platform for Mobile App Development



The proud boast of Codename One developers is, “Write once, run anywhere native mobile apps.” And it works exactly that way. You can use your existing Java skills to target iOS, Android and Windows devices with just one code base. Already, 120 million apps have been built on this platform!

In the current era of technology based products and applications, almost every business service is compatible with mobile devices. Mobile applications are everywhere, be it e-commerce, e-bidding, e-learning, e-reservations, discussion forums, e-banking or social media. Users are increasingly accessing the Internet on their mobile phones rather than on a PC or laptop. Hence there is immense scope for developing new mobile apps for multiple uses and devices.

There are a number of frameworks and platforms available to develop mobile phone apps. And the R&D teams of leading software development companies continue to explore this domain in order to come up with more dynamic products and services for the global market.

Table 1 gives a list of prominent mobile app development platforms and frameworks, which integrate the tools for user interface design and SDK for device-specific features with cross-platform compatibility modules.

Codename One: Free and open source software with SaaS (Software as a Service)

Codename One is among the prominent and unique device-independent platforms used for the development of cross-platform applications. It can be used to develop Java based applications that are deployed on any mobile device, irrespective of the base platform.

With Codename One, there is no need to write the mobile apps for different devices separately. The same code can be deployed with the same functionality and user interface in all the devices. The platform takes up very little memory, so applications developed with it can be executed on just 2MB of RAM, which is relatively low. This is a very significant feature because it means that apps developed on Codename One can run on virtually any smartphone.

As on June 30, 2015, there were 120 million apps installed that were built on Codename One, and 32,970 developers were using Codename One as per statistics given on the platform's portal.